MILITARY SPECIFICATION

AURAL PROTECTORS, SOUND

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types aural sound protectors that fit around the outer ear (pinna) designed primarily for personal hearing protection.

1.2 Classification. Aural protectors shall be of the following types as specified:

Type 1 - Metal headband
Type 2 - Dielectric headband

1.2.1 Military part number. The military part number shall consist of the letter M, the basic number of the specification and the type number as follows (see 6.2):

M38268 - 1
- 1: Type 1 or 2
- M38268: Document identification number

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions,deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Warner Robins Air Logistics Center (MMEDTA), AFLC. Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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SPECIFICATIONS

FEDERAL

PPP-B-636	Box. Fiberboard
PPP-P-291	Paperboard, Wrapping, Cushioning
UU-P-533	Paper, Wrapping, Tissue

MILITARY

MIL-P-116	Preservations, Methods of
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-36497	Spectacle Fronts and Temples, Cellulose Acetate

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by attributes
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-970	Standards and Specifications, Order of Preference for the selection of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity of as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ACOUSTICAL SOCIETY OF AMERICA

ANSI S12.6-1984 - Method for the Measurement of Real-Ear Attenuation of Hearing Protectors

(Copies may be obtained from the Standards Department, Acoustical Society of America, 335 East 45th Street, New York 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D3951 - Standard Practice for Commercial Packaging (DOD Adopted)
ASTM D149-81 - Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies, Test Method for

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The aural protectors furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Materials. Materials shall conform to the specifications and requirements specified herein. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970. Materials not specified shall be entirely suitable for the purpose intended (see 3.4, and 6.1).

3.2.1 Resistance to environment.

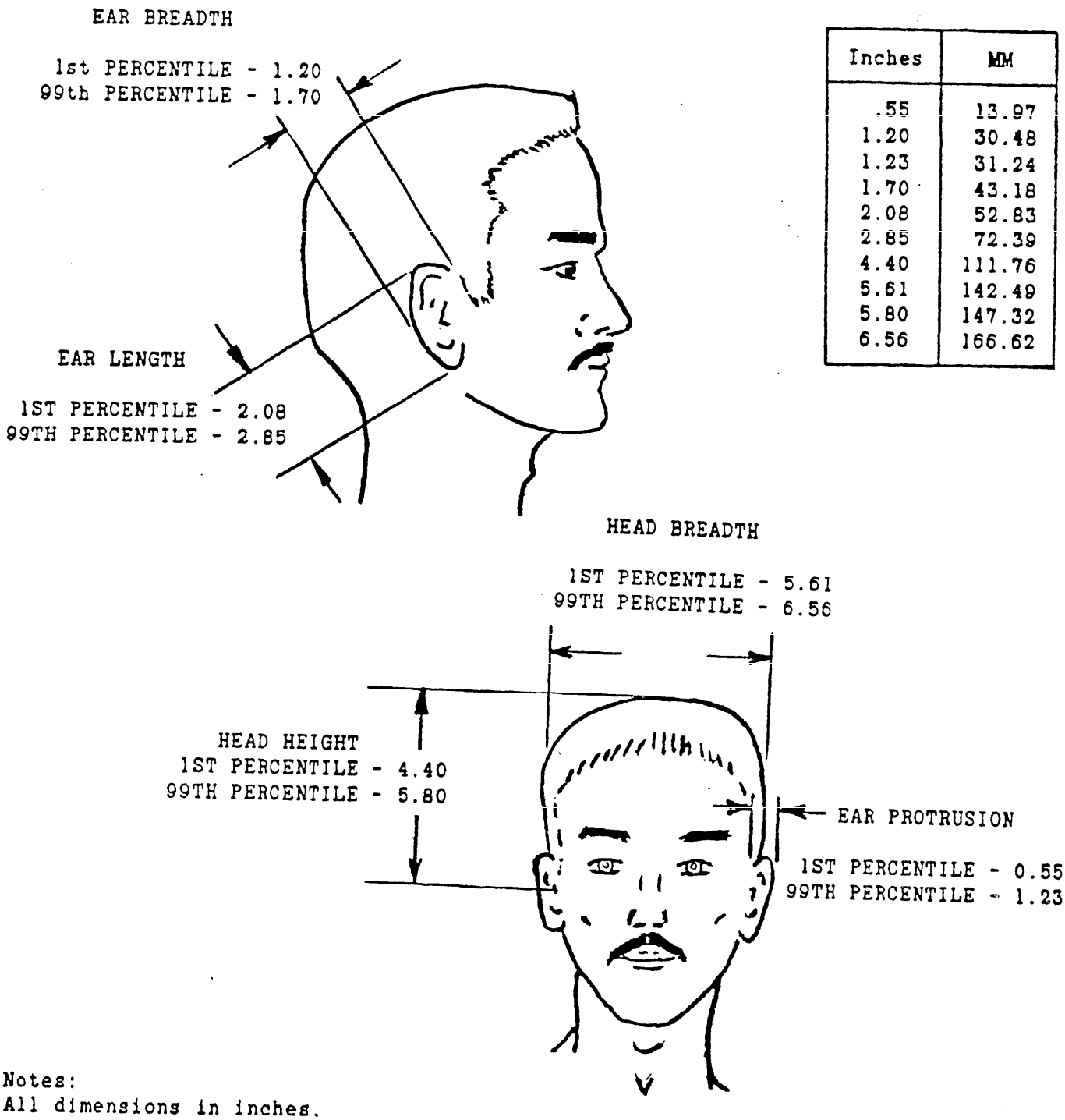
3.2.1.1 Protective treatment. Materials that are subject to corrosion in salt air or any other atmospheric conditions likely to occur during storage or usage shall be protected against such corrosion. The corrosion protection method used shall comply with the performance requirements of this specification and shall conform to the salt fog requirement, Method 509.1 of MIL-STD-810 (see 4.7.3). All aluminum parts shall be subjected to type II anodization in accordance with MIL-A-8625 (see 4.7.4). Any protective coating that will crack, chip or scale with age or extremes of atmospheric conditions shall not be used.

3.2.1.2 Temperature resistance. Aural protectors shall be capable of withstanding a temperature range of from -20°F (-29°C) to 150°F (65.5°C) when tested in accordance with 4.7.5.

3.2.2 Recycled materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements of this specification.

3.3 Design and construction. The aural protectors shall be designed to be comfortably worn on any size head and ear as specified in figure 1. It shall produce the sound attenuation specified in 3.4.3 when worn by all personnel within these measurements (including personnel wearing spectacles) who are subjected to high intensity sound. The aural protectors shall have a minimum number of parts consistent with reliability and shall be designed to function as one inseparable

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Inches	MM
.55	13.97
1.20	30.48
1.23	31.24
1.70	43.18
2.08	52.83
2.85	72.39
4.40	111.76
5.61	142.49
5.80	147.32
6.56	166.62

Notes:
All dimensions in inches.
Reference: Military Handbook Athropometry of U.S. Military Personnel, DOD-HDBK-743

FIGURE 1. Head and ear measurements

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unit (except for ear seals and fillers). Design shall permit assembly and disassembly with commercially available tools. The aural protectors shall consist of a suspension assembly and ear cup assembly. Headpads are not required to be a component in aural protectors utilizing one piece headbands. The aural protectors shall be designed to be worn either with the suspension over the head, back of the head, or under the chin.

3.3.1 Suspension assembly. The suspension assembly shall consist of a headband, support strap, head pad (with 2-piece headbands), and an attachment assembly.

3.3.1.1 Headband. The length of the headband shall be adjustable for ear cups to fit any size head as specified in figure 1. The tension of the headband shall not cause the ear cups to exert more than 4.5 pounds force against any specified width of the user's head, yet tension shall be sufficient to ensure the sound-attenuating qualities are not sacrificed. The headband shall be constructed of a material not less than one-half inch wide and shall meet the flexing requirements specified in 3.3.1.2 and 3.4.1 when tested as specified in 4.7.7. The type 2 headband shall have a minimum dielectric strength of 400 volts per mil as tested in accordance with ASTM D149-81. Dielectric strength of the headband material shall be determined by the testing as specified in 4.7.6 or certification of results as tested in accordance with ASTM D149-81. If certifications are to be to be presented, two certifications are required: one shall be signed by a responsible officer of the material manufacturer referencing the batch or lot number. The second shall be provided by a responsible officer or the aural protectors manufacturer attesting the headbands are made from this lot of certified material.

3.3.1.2 Reliability. The aural protectors headband shall have a mean-cycles-between-failures of at least 2,174 cycles at a confidence level of 90 percent when flexed as specified in 4.7.7 and 4.7.8.

3.3.1.3 Head pad. The material shall be a type that will not readily absorb moisture. For those aural protectors utilizing head pads, the head pad will provide cushioning between the headband and the head. The thickness of the pad shall provide comfort and prevent the headband from contacting the head. The material used for the head pad shall not harden to a point of discomfort and not become sticky when tested as specified in 4.7.5.

3.3.1.4 Support strap. The aural protectors shall be equipped with an adjustable support strap. The support strap shall be attached to the aural protectors and shall consist of a minimum one-half inch wide ribbon or band of nylon (or equivalent material of equal strength, flexibility and perspiration resistance) with adjustment mechanism to provide stabilization of the aural protectors when worn in other than the over-the-head position (see 4.7.12).

3.3.1.5 Attachment assemblies. The attachment assembly shall be designed to join the ear cups to the headband. The attachment assembly shall be designed to allow sufficient adjustment between the headband and ear cups so that the aural protectors provide comfort and proper sound attenuation in all positions (see 3.4.4). The ear cups shall remain attached to the headband when tested in accordance with 4.7.11.

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3.3.2 Ear cup assembly. The ear cup assembly shall consist of ear cups, fillers (at the manufacturer's option) and seals.

3.3.2.1 Ear cups. Each aural protector shall have two ear cups. The ear cups shall be free from sharp corners. The ear cup openings shall be large enough to accommodate the 1st and 99th percentile user (see figure 1). The size of the ear cups shall be the minimum necessary to provide the attenuation specified in 3.4.3. If the design of the ear cups necessitates differentiation between the front and back, and/or the top and bottom of the ear cups for proper use, permanent designation markings shall be provided. Ear cups constructed of more than one piece shall be permanently joined together. Ear cups produced by the same manufacturer shall be interchange- able with each of the same model.

3.3.2.2 Seal set. A seal set shall consist of two ear cup seals and be designed as replaceable items not cemented to the ear cups; however the seals shall remain firmly attached to the ear cups during normal use of the aural protectors. One additional set of seals shall be packed in the container with each aural protector. The seals shall be designed to conform to various head contours while the user is wearing spectacles. The seals shall remain attached to the ear cups during normal usage of the aural protectors. Material coming in contact with the skin of the wearer shall be of the closed-cell type and shall not be treated in any manner that might cause skin irritation. The thickness of the seal shall provide comfort and prevent the ear cup from contacting the head as specified in 4.7.13. The material used for the seal shall not harden to a point of discomfort and not become sticky when tested as specified in 4.7.5. The material shall be a type that will not readily absorb moisture. The use of polyvinyl chloride is prohibited.

3.3.2.3 Fillers. The fillers shall be designed so that they can be removed from the ear cup cavities and replaced without damage to the fillers or ear cup. The filler material shall not be glued to the ear cups and shall be material which will assure the sound-attenuating requirements specified in 3.4.3 for the service life of the aural protector. The material used shall not cause skin irritation.

3.4 Performance.

3.4.1 Headband flexibility. The headband shall be capable of being extended to the 99th percentile height and opened to 7.1 inches and shall not deform nor weaken when tested as specified in 4.7.7.

3.4.2 Shock. The aural protectors shall be shock tested and shall function as specified in 4.7.9.

3.4.3 Sound attenuation. Individual and Group sound attenuation of the aural protectors shall meet the requirements of table I when tested and measured in accordance with the American National Standards Institute, Method for the Measurement of Real-Ear Attenuation of Hearing Protectors S12.6-1984. Attenuation values should be rounded to the nearest whole number. The group attenuation shall meet the sum of the individual attenuations measured for the test frequencies of that group. Measurements shall be taken in three headband positions while wearing military issue spectacle frames as specified in MIL-S-36497: a. over-the-head b. behind-the-head and c. under-the-chin. A 3 dB reduction both in each group and in

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individual frequency values should be applied for under-the-chin and behind-the-head headband positions.

3.4.4 Adjustments. The amount of vertical adjustment shall be a minimum necessary to accommodate the 1st to 99th percentile head height as specified in figure 1. The ear cup assembly, when attached to the suspension assembly shall be sufficiently adjustable to ensure easy fitting and holding in each position when tested as specified in 4.7.13.

TABLE I.- Sound attenuation

Group	Minimum group attenuation (decibels)	Frequencies (Hz)	Minimum individual attenuation (decibels)
A	25	125 250	
B	157	500 1,000 2,000 3,000 4,000	25 30 32 35 35
C	58	6,000 8,000	28 28

3.5 Weight. The complete aural protector with strap shall not weigh greater than 14 ounces (see 4.7.1).

3.6 Marking. The aural protectors shall be marked in accordance with MIL-STD-130.

3.7 Workmanship. Workmanship and finish shall be of high quality, in accordance with the best commercial practice. Protective coatings shall cover uniformly and shall be free of pitting, surface cracks, 'orange peel', peeling, chipping, scale, and foreign bodies. Unprotected surfaces of the aural protectors shall be free of striations, pitting, surface cracks, grease and rust. Group A inspections shall be performed on the aural protectors for defects that might affect serviceability or appearance (see 4.5.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. Qualification inspection, Group B & C (see 4.3 and 4.6).

b. Verification of qualification (see 4.4).

c. Quality conformance inspection, group A (see 4.5).

4.3 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the government (see 6.3) on sample units produced with equipment and procedures normally used in production. Qualification inspection shall consist of groups A,B & C inspections.

4.3.1 Sample size tested by the contractor. The manufacturer shall subject four complete aural protectors, each with one additional set of seals, to all the inspections specified in 4.3. Any sample failing any examination in groups A,B or C shall be cause for failure to qualify.

4.3.2 Test report. After the tests have been completed, the manufacturer shall prepare a qualification test report in accordance with MIL-STD-831 and shall furnish five copies of the report to the qualifying activity whether the samples pass or fail the inspections.

4.4 Verification of qualification. To retain qualification, the contractor shall forward a report at 6-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the tests performed for inspection of product for delivery, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

b. A summary of the results of tests performed for periodic inspection, including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 6-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

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Failure to submit the report within 30 days after the end of each 6-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 6-month period that the inspection data indicates failure of the qualified product to meet the requirements of this inspection.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit his qualified products to testing in accordance with the qualification inspection requirements and the reason for no production.

4.5 Quality conformance inspection. Quality conformance inspections are as follows:

Group A inspection (see 4.5.2).

Packaging inspection (see 4.8).

The sample size, inspection levels, acceptance criteria, and inspections required for the aural protectors shall be as specified in table II.

TABLE II. Quality conformance sampling plan.

Inspection	Method paragraph	Sample size 1/	Acceptance criteria 1/
Group A	4.5.2	Inspection level II	Major defects - AQL is 1.5 Minor defects - AQL 2.5
Packaging	4.8	Inspection level II	All defects - AQL is 2.5

1/ see MIL-STD -105.

4.5.1 Inspection lot. A lot size shall be expressed in units of not more than 500 aural protectors. The sample unit shall be one aural protector and one extra set of seals.

4.5.2 Group A inspection. Each aural protector shall be inspected as specified in table III and examined visually to determine conformance to this specification. Acceptance shall be determined by the number and severity of defects in table IV.

4.5.2.1 Sampling plan. The sample size, acceptance criteria, tests and examinations required for the aural protectors shall be as specified in table II.

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4.6 Group B and C inspections. Each aural protector shall be submitted to Group B and C inspections as in the order shown and specified in table V. They shall be performed on sample units that have been subjected to and have passed the group A inspections of table II.

TABLE III. Group A - quality conformance tests.

Inspection	Requirement paragraph	Test paragraph
Dimensions	3.3	4.7.2
Weight	3.5	4.7.1
Design and construction	3.3	4.5.2 & 4.6
Visual inspection	3.7	4.5.2
Marking	3.6	Table IV
Support strap	3.3.1.4	4.7.12

TABLE IV. Classification of defects.

Defects	Major	Minor
Any characteristic not in accordance with specified requirements.	X	
Dimensions not as specified.	X	
Any hole, cut, tear, rip, brittle area or crack in the headpad, dome seal or dome filler.	X	
Any component missing, cracked or broken, malformed or not functioning properly.	X	
Any components improperly assembled molded or brazed.	X	
Any sharp edges or burrs and damaged or stripped threads.	X	
Lack of protective coating (see 3.2.1.1). Defective finish or surface as specified in 3.7.	X	
Scratched, scuffed, or abraded areas on dome surface.		X
Marking missing, incorrect, incomplete, illegible, misplaced or not as specified in 3.6.		X
Seals not detachable or not firmly attached after replacing in accordance with 3.3.2.2.	X	
Fillers not in accordance with 3.3.2.3	X	

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4.7 Methods of inspection.

4.7.1 Weight. The aural protectors shall be weighed and shall not exceed 14 ounces.

TABLE V. Qualification tests.

Inspection	Requirement paragraph	Test paragraph
Group B		
Temperature resistance	3.2.1.2, 3.3.2.2	4.7.5, 4.7.7
Headband flexibility	3.3.1.1, 3.3.1.2, 3.4.1	4.7.7
Headband dielectric strength	3.3.1.1	4.7.6
Head pad	3.3.1.3	4.7.5
Ear cups	3.3.2.1	4.7.13
Attachment assembly retention	3.3.1.5	4.7.11
Seals	3.3.2.2	4.7.3, 4.7.5
Shock	3.4.2	4.7.9
Sound attenuation	3.4.3	4.7.10
Adjustments	3.4.4	4.7.13
Group C		
Protective treatment	3.2.1.1	4.7.3, 4.7.4 4.7.5
Reliability demonstration	3.3.1.2	4.7.8

4.7.2 Dimensions. The aural protectors shall be thoroughly checked dimensionally to determine conformance to 3.3 and figure 1.

4.7.3 Resistance to environment. Each aural protector shall be subjected to a 48 hour test, as specified in 3.2.1.1 using a wet, dense salt spray to determine the corrosive resistance of the aural protectors to the effects of salt atmosphere. Upon completion of this test, there shall be no evidence of damage to any parts of the aural protector and it shall be rinsed and allowed to dry for 48 hours prior to examination. Aural protectors shall show no evidence of excessive salt deposits or corrosion which affects performance as specified herein.

4.7.4 Anodization. All aluminum parts of the aural protectors shall be tested and meet the requirements for type II anodic coatings as specified in MIL-A-8625.

4.7.5 Temperature resistance. The aural protectors shall be subjected to -20°F (-29°C) for a minimum of three hours, removed and inspected for compliance with 3.3.1.3. and 3.3.2.2, then allowed to return to ambient temperature and evaluated. Another device shall be subjected to dry circulating air at a temperature of 150°F (65.5°C) for a minimum of 48 hours, removed and inspected for compliance with 3.3.1.3, and 3.3.2.2, then allowed to return to ambient conditions and evaluated. Failure shall be signs of splitting, hardening, stickiness, permanent deformation, change in size, suspension system breaking, or corrosive protective coating not complying with 3.2.1.1.

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4.7.6 Type 2 Headband dielectric test. Testing shall be in accordance with ASTM D149-81 using method A voltage application with electrode type 3 on a 4 centimeter square of the same thickness material as the headband. The material shall be submerged in the recommended oil at 73° ±2°C (22.8° ±1.1°C) during the test.

4.7.7 Headband flexibility. With the aural protector assembly extended to the 99th percentile height and placed on a block that will provide 6.1 inches of space between ear seals, measure the force exerted against the block. Attach the aural protector assembly to a suitable flexing device and flex the ear cup assemblies to a distance 1 inch greater (6.1 to 7.1 inches) 5,000 cycles (a cycle is defined as one use of the aural protector which involves two flexures of the headband: outward then returned). The force exerted on the block after flexing shall be 90% or greater of the force prior to flexing. This test shall be conducted with the aural protector assembly stabilized at -25°F (-31.6°C) (see 4.7.8).

4.7.8 Headband reliability demonstration. Satisfactory completion of all the tests listed under 4.7 without failure demonstrates the requirement of 3.3.1.2 (a reliability of 2,174 cycles at a confidence level of 90 percent). If no failure occurs during the 5,000 cycles (10,000 flexures) of the aural protector in 4.7.7, this test is satisfactory; one failure requires another aural protector be substituted and the testing continue through 8,457 cycles, as outlined in table VI. Over three failures of this test shall be cause for rejecting qualification.

TABLE VI. - Rejection criteria

Failures	Total test in cycles (2 flexures)
0	5,000
1	8,457
2	11,522
3	14,566
Over 3	1/

1/ aural protector will be rejected and qualification test fails.

4.7.9 Shock. Three aural protectors shall be cooled to a temperature of -20°F (-29°C) for at least 3 hours. After cooling, each aural protector shall be removed from the chamber, and within 30 seconds dropped onto a concrete floor from a height of 6 feet, to strike the floor in the following positions:

a. First aural protector; headband up.
b. Second aural protector; headband down.
c. Third aural protector; one side down.

After the drops the suspension shall adjust, rotate, or slide as intended, and there shall be no separated, broken or cracked components. After this shock test the aural protectors shall be subjected to the tests of 4.7.10. Ear seals may be replaced prior to the tests of 4.7.10.

4.7.10 Sound attenuation. The aural protectors shall be tested as required in 3.4.3. Attenuation shall be tested at all frequencies listed in table I.

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4.7.11 Attachment assembly retention. The aural protector assembly shall be installed on a fixture that will hold the headband at the center (two piece headbands shall be extended fully to maximum length). With the ear cups resting on the attachment assembly, apply a 25 pound load to each ear cup assembly for five minutes. Separation of the attachment assembly or the headband shall be cause for rejection.

4.7.12 Support strap. The support strap width shall be measured to verify adherence to 3.3.1.4. After the attachment assembly test (4.7.11), secure the aural protector on the fixture, apply a 30 pound load to the center of the support strap for 15 minutes thereby applying 15 pounds pressure at each connection. Separation at any point of the strap shall cause rejection. The strap material shall be tested for pH value and shall be within the range of 5 to 9, if any reaction is produced from the sample.

4.7.12.1 Tearing strength. The apparatus for this test shall consist of grip clamps for anchoring and for load application of a type which shall preclude failure of the specimens at or in the grips. Three specimen straps taken from the sample aural protector shall be tested. The clamps shall hold the specimen uniformly across the one-half inch dimension. One end of the specimen shall be anchored in a clamp and a load of 15 pounds shall be applied at the other end for one hour at 65% ±2% relative humidity at a temperature of 70°F ±2°F. The anchoring and load application clamps shall be applied to the specimen. The load shall be applied uniformly and without twist. Gage marks should be made at the point where the gage is initially placed on the specimen to assist in determining slippage. At the end of the test period, the tension shall be removed from the specimen, and the specimen examined. The strap length shall stretch no more than one inch, tear, or indicate possible tearing. Any of the specimen straps failing this test shall cause failure to qualify.

4.7.13 Adjustments. The attachment assembly shall be cycled from behind the head to over the head, to under the chin, and returned 500 times. The ear cups will be adjusted to fit evenly against the user's head at each position yet not contact the head (see 3.3.2.2 and 3.4.4). Upon completion, the friction required to maintain the attachment assembly in a horizontal behind-the-head position shall not be reduced to cause it to freely rotate by its own weight.

4.8 Inspection of packaging. One sample unit pack shall be selected at random from each lot of not more than 500 that are ready for shipment (see table II for sampling plan). Each item shall be examined for the defects specified in table VII to determine conformance with requirements of section 5.

5. PACKAGING

5.1 Preservation and packing. Preservation and packing shall be level A or C as specified (see 6.2). The quantity per unit pack shall be one aural protector and one additional set of seals. A unit pack shall be so designed and constructed that it shall:

a. prevent ear cups from pressing together, causing the seals to be pinched or distorted,

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TABLE VII. Packaging inspection.

Item	Defects
Markings	Omitted, incorrect, illegible; improper size, location, sequence or method of application.
Materials	Any component missing; any component damaged, affecting serviceability. Additional set of seals missing.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping or inadequate stapling; bulging or distortion of containers.
Exterior and interior weight or content	Number of intermediate packages is more or less than required; gross of net weight exceeds requirements.

b. insure plastic headbands of aural protectors will remain in a normal unstressed position to effect minimum damage to the unit pack during shipment and storage in the shipping container,
c. contain an additional set of seals packed in the unit pack in such a manner to prevent distorting the normal shape and
d. allow subsequent handling with minimum damage.

5.1.1 Level A. Preservation and packing shall be in accordance with Method III of MIL-P-116. Each aural protector and set of seals shall be wrapped with material conforming to PPP-P-291 or UU-P-553 and shall be packed in a unit container conforming to CF, Class - domestic.

5.1.2 Level C. Aural protectors and additional set of seals shall be packed in accordance with ASTM D3951 in a manner which will ensure adequate protection against deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A,B, or C as specified (see 6.2).

5.2.1 Level A. The aural protectors shall be packed in overseas-type containers conforming to PPP-B-636. The containers shall be of minimum weight and cube, uniform size and shape, and shall contain like quantities. The gross weight of the containers shall be governed by the container specification. Closure of the containers shall be in accordance with the specification and appendix thereto.

5.2.2 Level B. Except that containers shall be of the domestic type and closure shall be in accordance with the requirements of domestic containers, the aural protectors shall be packed as specified in 5.2.1.

5.2.3 Level C. The packages that require overpacking for acceptance by the carrier shall be packed on shipping containers which will ensure acceptance by common carrier at the lowest rates and safe delivery at destination. Packing shall

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comply with ASTM D3951 and the Uniform Freight or National Motor Freight Classification Rules and Regulations, as applicable.

6. NOTES

6.1 Intended use. The aural protectors covered by this specification is intended to be used by military personnel required to be in areas where, due to the intensity of sound, ear protection is required. They are not intended to be used with a military helmet or chemical & biological protection gear. Type 1 aural protectors with metal headbands are for general use. Type 2 aural protectors with dielectric headbands protect electrical and electronic equipment from damage and personnel from injury.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification and the military part number.
b. Selection of applicable levels of packing, preservation and packaging (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification (Copies of 'Provisions Governing Qualification SD-6' may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120). The activity responsible for the Qualified Products List is the Warner Robins Air Material Area, attention WR-ALC/MMVRG, Robins AFB, GA 31098-5609, and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key word) listing.

Attenuator, sound	Mickey mouse ears
Aural protector	Noise suppressors
Earmuffs	Protector, hearing
Head sets	Sound attenuator
Hearing protector	Suppressor, noise

6.5 Changes from the previous amendment. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the change.

Custodian:
Army - EA
Navy - MS
Air Force - 84

Preparing activity:
Air Force - 84

Agent:
DLA - GS

(Project 4240-0509)

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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